Kickstart Mini-Challenge



Mini-Challenge Task

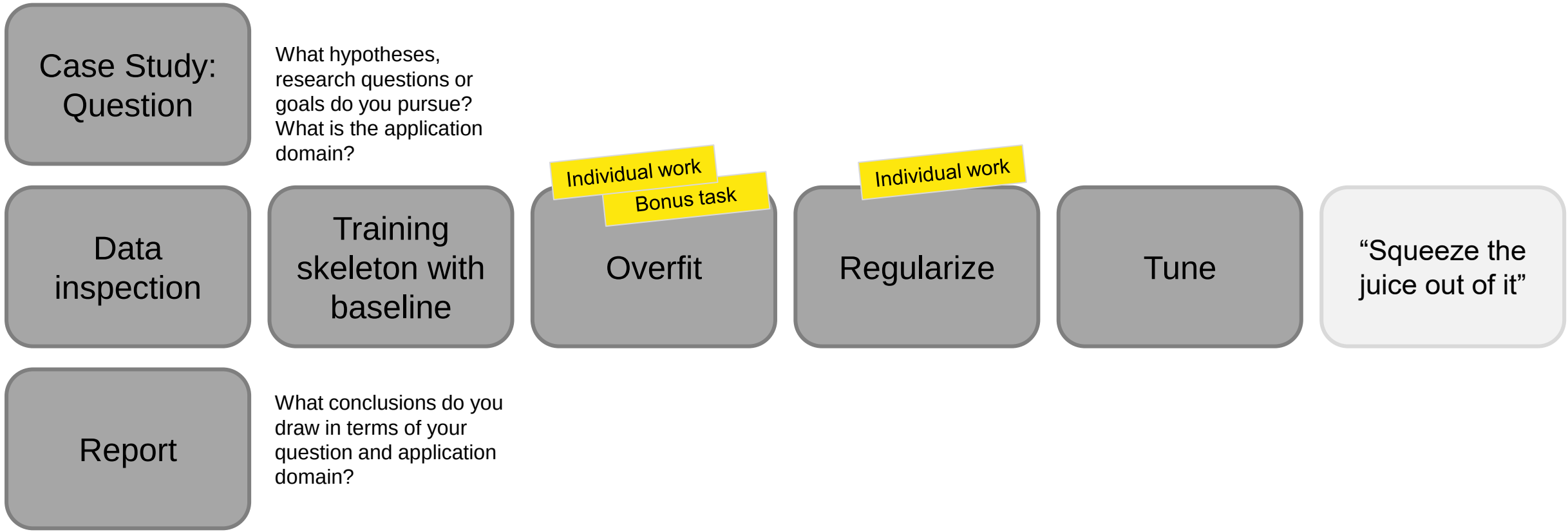
Choose a question from the areas of image or signal processing that you would like to work on in this mini-challenge. The problem should be solvable by object recognition, image segmentation, instance segmentation or signal prediction using deep learning.

The programming language can be chosen freely.

The solution procedure of the mini-challenge is based on the articles:

- How to Avoid Machine Learning Pitfalls: A Guide for Academic Researchers [Lon:23]
- A Recipe for Training Neural Networks [Kar:19]

Solution Approach



Don’t be a hero -> Less is More

If the model does not learn, do not continue until this can be solved (otherwise the work is insufficient)
If a model does not give results, there are no points
Increase complexity gradually (first only 1 class, then several, 1 model, then several)
Find a good and simple data set. If you can't cope with the first one, switch
I can't retrain the whole model - > i.e. you have to demonstrate to me where it works and where it doesn't.

Object Detection



Two-stage region proposals, e.g.:
- R-CNN [2013], Fast R-CNN [2015], Faster R-CNN [2015]
- Mask R-CNN [2017]
- G-R-CNN [2021]
- …

One-stage detectors, e.g.:
- YOLO [2015], aktuell YOLO V8 [2023]
- Single Shot Detector (SSD) [2015]
- CenterNet [2019] (keypoints)
- …

(Vision transformers – officially not part of this module, but may be chosen for mini-challenge)

Semantic Segmentation

- U-Net [2015] and variations
- SegNet [2015]
- DeepLabv3 / DeepLabv3+ [2018]
- Pyramid Scene Parsing Network (PSPNet) [2016]
- Masked R-CNN [2017]
- (Vision transformers – officially not part of this module, but may be chosen for mini-challenge)
- …



Semantic Segmentation



Instance Segmentation

Newly Released Segment Anything Model (SAM)

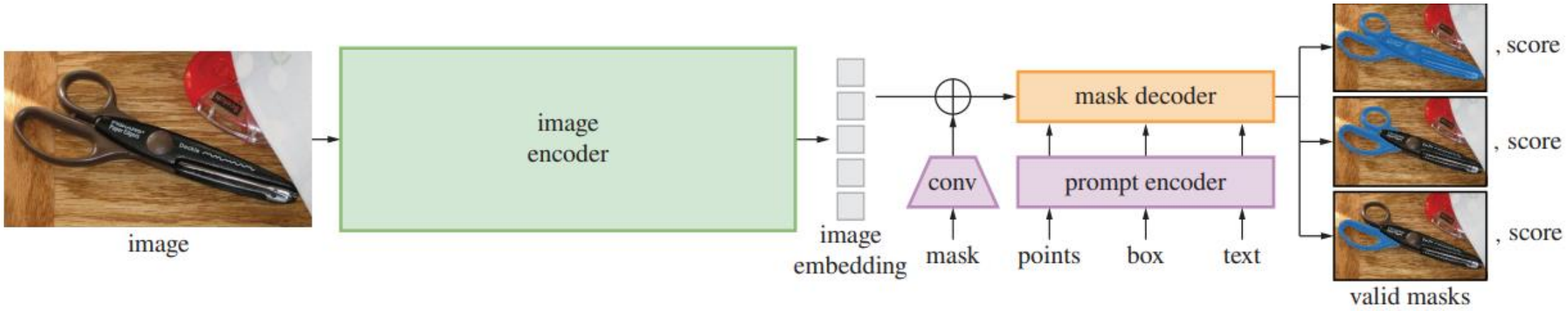


Figure 4: Segment Anything Model (SAM) overview. A heavyweight image encoder outputs an image embedding that can then be efficiently queried by a variety of input prompts to produce object masks at amortized real-time speed. For ambiguous prompts corresponding to more than one object, SAM can output multiple valid masks and associated confidence scores.

Instance Segmentation: Mask-R-CNN

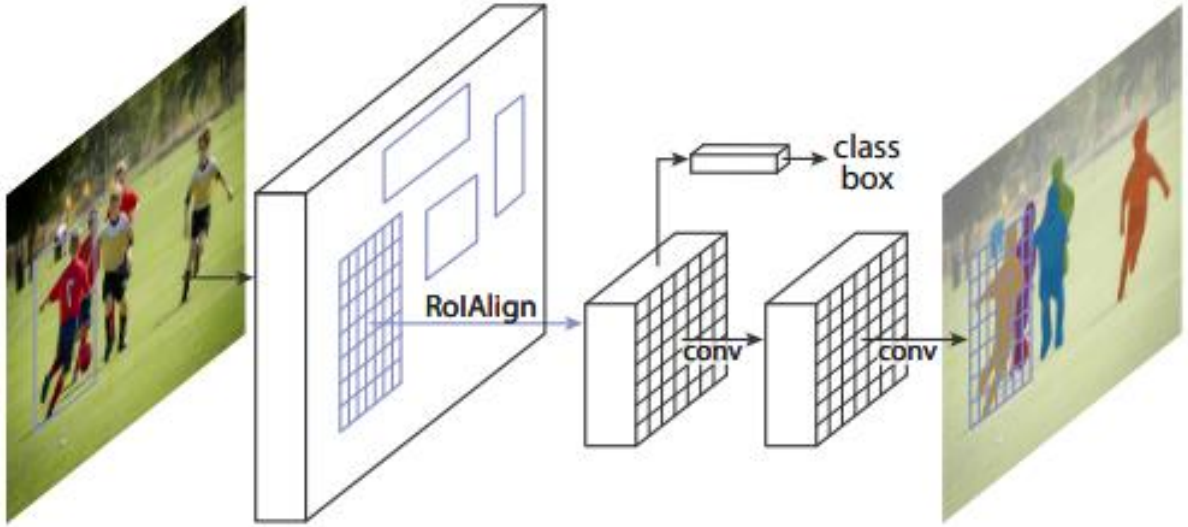


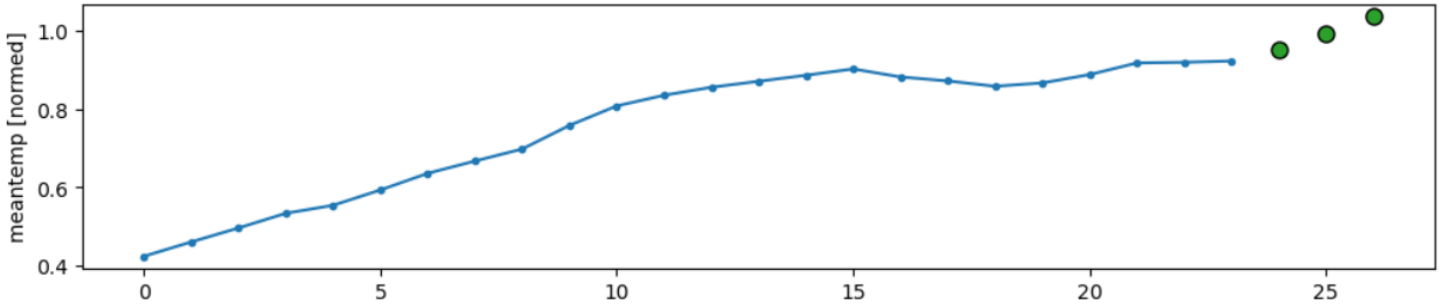
Figure 1. The **Mask R-CNN** framework for instance segmentation.

When to Use What Method?

- Classification: objects that have no physical presence, e.g., “sunny”, “bright”, or “skewed”
- Object detection is very good at:
  - Detecting objects that take up between 2% and 60% of an image’s area
  - Detecting objects with clear boundaries
  - Detecting clusters of objects as 1 item
  - Localizing objects at high speed (>15fps)
- Semantic segmentation: objects that have no clear boundaries at different angles, e.g., sky
- Instance segmentation: objects that are elongated: model bias because of scarcity of objects
- Instance segmentation: objects that are often occluded: occlusion is handled far better in two-stage detection networks than one-shot approaches

Signal Prediction

- RNNs
- LSTMs
- GRUs
- CNNs
- (Transformers – officially not part of this module, but may be chosen for mini-challenge)
- …



Signal Prediction

Classification or regression?

- How accurate does the prediction need to be? How accurate can the prediction be expected?
- Get a confidence / probability from the regression task
- Try both approaches and compare results
- Apply classification in post-processing, e.g., using thresholding

Using Images Instead of Signals

- Line plots et al. as images to detect patterns
- Spectogramms

Case Studies

1. Choose a case study that personally motivates you (interest, hobby, working field, …)
2. Get inspired, e.g.:
   - https://www.v7labs.com/blog/computer-vision-applications
   - https://www.v7labs.com/blog/computer-vision-project-ideas
3. Look for good available datasets including labels: how many samples, classes, labels, class imbalances etc.
4. Formulate research question explicitly and clearly

Staring at data including labels and looking for patterns

Working in a application goal-oriented manner

What makes a good dataset for the case study?

- Does the data set fit the question? How to formulate the question?

- Number of samples: a few hundred to a few thousand – depending on the number of object classes
- Number of property classes: 2-10
- Image sizes: guide size 500x500 pixels see also model specifications of the selected architectures
- Object classes, distributions, statistics, imbalances, bias, ...? Consider when training. E.g. weighted loss
- Variability of data? Does this target population/target data correspond?
- Labels are available and validated or look ok
- Can objects be easily identified/separated by eye
- What do experts say about it? Other students, friends, work colleagues, ... What does the literature say about it?

- Get qualitative and quantitative sense of data, see [Kar:19] and [Lon:23]

Public Dataset Sources

General dataset sources:

- https://www.kaggle.com/datasets
- https://datasetsearch.research.google.com

- Vision datasets:
  - https://www.v7labs.com/open-datasets
  - https://roboflow.com/universe
  - https://lila.science/otherdatasets: labeled information library of Alexandria (LILA): biology and conservation
  - https://www.v7labs.com/blog/computer-vision-datasets
  - https://pytorch.org/vision/stable/datasets.html
  - …

- Labeling tools:
  - https://supervisely.com/ (free community account)
  - https://www.v7labs.com/image-annotation (free edu account)
  - Roboflow
  - …

Useful Tools

- Helpful concepts:
  - Images:
    - Bounding boxes: get enclosing rectangle of a region
    - Convex hull / concave hull: convert polygon into pixel-wise labels
    - See example notebook to visualize image labels (or deep dive notebooks)

- Libraries:
  - Images: Pillow (PIL), OpenCV
  - Signals: Librosa (music / audio), SciPy

- Data augmentation: https://neptune.ai/blog/data-augmentation-in-python

- Image viewer locally: https://www.irfanview.com/

- MLOps tools: WandB, TensorBoard, Neptune.ai, MLFlow, …

Much success with your dlbs mini-challenge!

